Tutoring 1 - European Economic Policy

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Exchange rate and competitiveness

E: nominal bilateral exchange rate

It tells the price of a foreign currency in terms of 1 unit of national currency

We use *E* when we want to compare the price of the same good i sold in two countries:

P^i_{US} *vs.* $P^i_{EZ} * E$

P^i_{EZ} *vs* P^i_{US} / E

Law of one price

If $P^i_{EZ} * E = P^i_{US}$, then also $P^i_{US} / E = P^i_{EZ}$

If the law of one price holds for all goods in a given basket, there is:

Purchasing Power Parity (PPP)

All countries' price levels are equal when measured in the same currency.
It means that I can buy the same good in both countries with the same amount of money. Let's check

Big Mac Index (July 2021)

Let's review:

Price Big Mac in EZ: €4.27
Price Big Mac in USA: $5.65
Market exchange rate: 1.18

What is the exchange rate necessary to obtain PPP?
Is it the same as the market exchange rate?
What are the implications?

From PPP to real exchange rate

If PPP holds, then $E = P_{US} / P_{EZ}$
and also $P_{EZ}\, E / P_{US} = 1$

But PPP performs poorly in the real world

Deviations from PPP describe the actual exchange rate:

$E_R = P_{EZ}\, E / P_{US}$

From real exchange rate to measuring competitiveness

$E_R = P_{EZ}\, E / P_{US}$

If PPP holds, $E_R = 1$. In the real world, it can be:

$E_R > 1$ --> EZ goods are more expensive than US goods

$E_R < 1$ --> EZ goods are less expensive than US goods

What does it mean for competitiveness?

Competitiveness rate of change

Starting from $E_R = P_{EZ}\, E / P_{US}$, we are interested in change over time

From levels to rates of growth: how do we do it?

Exercise A

Exchange rates:

$E_{t-1} = E_{\text{Sep 2024}} = 1.11618$ USD/EUR

$E_{t} = E_{\text{Sep 2025}} = 1.1736$ USD/EUR

Base price indices: $P^{EZ}_{t-1} = 100$ $P^{US}_{t-1} = 100$

Annual inflation rates (12-month to Aug 2025):

$p^{\cdot EZ} = 2.0\%$ $p^{\cdot US} = 2.9\%$

Comment on change in EZ-USA bilateral competitiveness over time by computing real exchange rates:

1. In levels

2. Using log / growth rate approximation

Exercise B

Exchange rates:

E_{t-1} = 1.20 USD/EUR

E_{t} = 1.10 USD/EUR

Base price indices: P^{EZ}_{t-1}=100 P^{US}_{t-1} =110

Annual inflation rates:

$\dot{p}^{EZ}$=1% $\dot{p}^{US}$=4%